

# **Exhibit CC-110**

# Exhibit 7

# **A Dive into DDR5:** *An Engineer's Guide to Simulating and Validating the Latest Generation of DDR*

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Sept. 22, 2021

# Meet the Presenter

- Stephen Newberry:
  - Signal Integrity Engineer at Astranis
  - B.S. in ECE from Rutgers
  - Former U.S. Coast Guard Electrician's Mate
  - Husband and father of 4

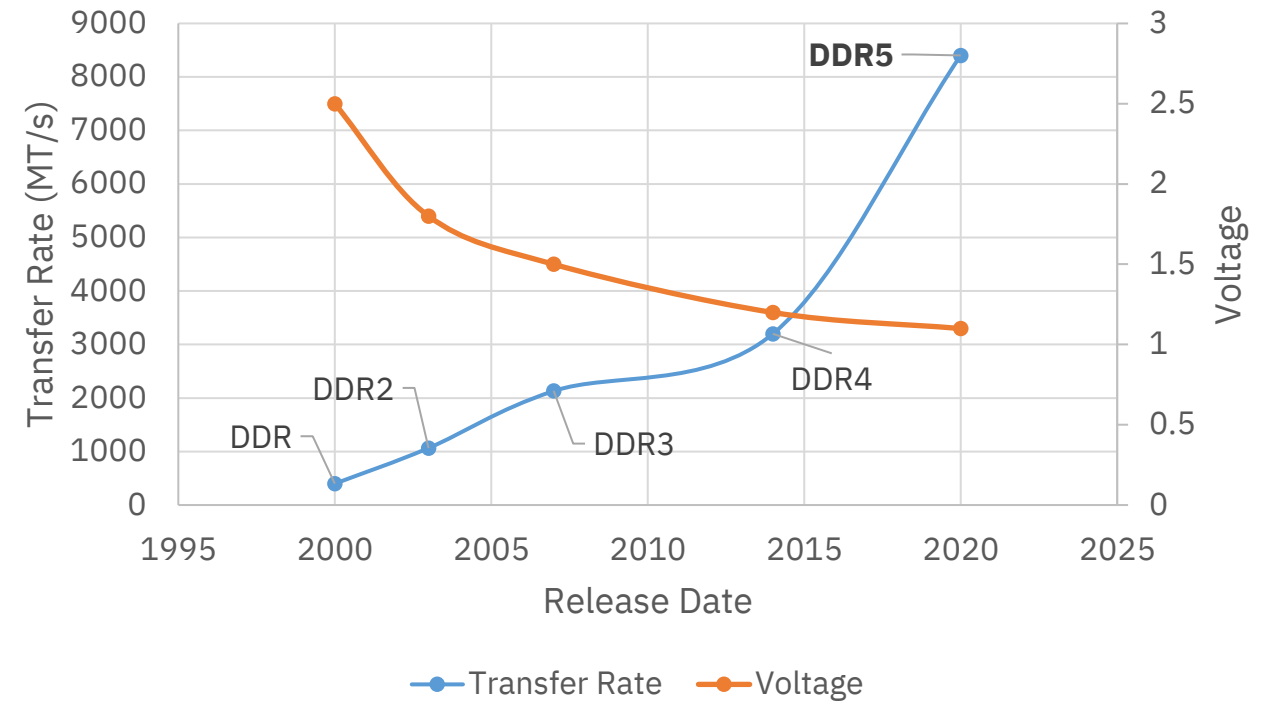


ASTRANIS

# Introduction

- This overview has **two goals**:
  - Provide **basic info** about DDR5 changes
  - Introduce simulation with **IBIS-AMI** and **DFE**
- Intended audience:
  - **Familiar** with prior DDR specifications
  - Want to see the **major updates** in DDR5
  - Jump into **simulation**
- DDR4 is fast
  - DDR5 is **blazing**
  - Tables in JESD79-5 have placeholder text to **8400 MT/s!**
- Many clever features are employed to gain functionality
  - **Better** but not significantly **different** – basic architecture unchanged
  - **Not** (yet) using **differential** signaling for data
  - Package sizes and **pinouts** are similar
  - Advances require **alternate approach** to simulation and analysis

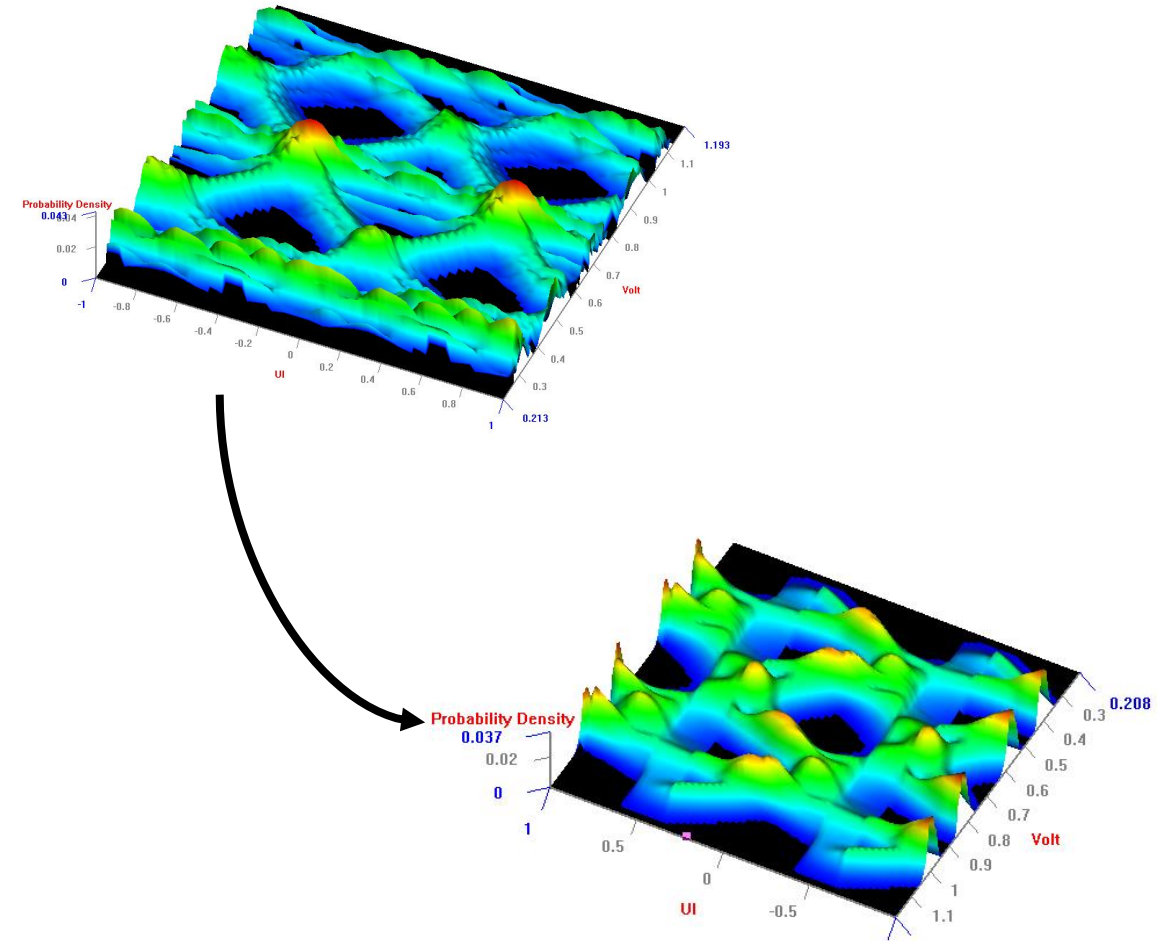
DDR Speed and Voltage over Time



Data from Wikipedia: [https://en.wikipedia.org/wiki/DDR\\_SDRAM](https://en.wikipedia.org/wiki/DDR_SDRAM)

# Outline

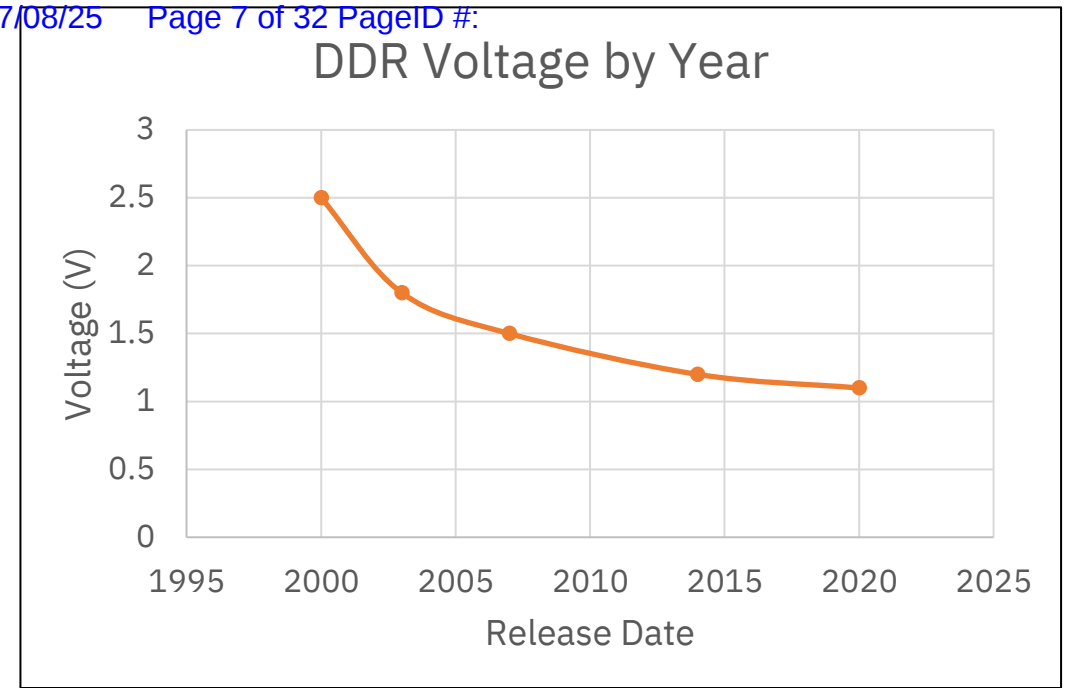
1. Major Changes from DDR4
  1. Reduced VDD/VDDQ
  2. CA Bus: Non-Specific Signal Names
  3. CA Bus: Internal Vref
  4. On-Die Termination Changes
2. New Features in DDR5
  1. VRM on DIMM
  2. Duty Cycle Adjuster
  3. On-Die ECC
  4. Decision Feedback Equalization
3. Simulation Example
  1. Full byte lane with DFE



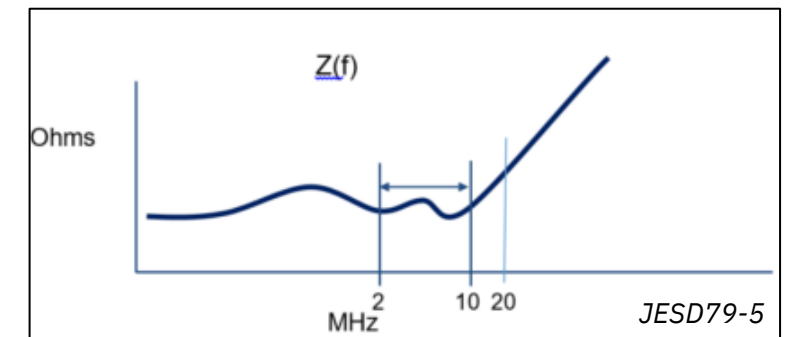
# 1. Major Changes from DDR4

# VDD/VDDQ to 1.1V

- DDR4 was 1.2V
- Sequence: VPP then VDD/VDDQ
  - Then RESET\_n
- DC Operating Range: +6%/-3%
  - On all VDD/VDDQ/VPP
- PDN Impedance preliminary specification:
  - 10 mΩ from 2-10 MHz
  - 20 mΩ at 20 MHz



Data from Wikipedia: [https://en.wikipedia.org/wiki/DDR\\_SDRAM](https://en.wikipedia.org/wiki/DDR_SDRAM)



# CA Bus: Non-Specific Signal Names

- DDR4 used specific pins: ACT\_n, RAS\_n, CAS\_n, etc.
- DDR5 uses generic bus: CA[0..13]
- No pin-swapping allowed
  - Commands may be multi-cycle
  - But be aware of MIR pin (mirror) for clamshell configuration
    - Internally swaps even CA pin with next highest odd CA
- Biggest impact to the designer?
  - Easier schematic capture: simple bus connection
- Fewer CA pins mean more GND pins
  - Improved return path = better signal integrity

DDR5 command encoding<sup>[22]</sup><sup>[final standard verification needed]</sup>

| Command                         | $\overline{CS}$ | Command/address (CA) bits |              |          |   |   |                 |      |            |                 |     |             |              |    |    |  |
|---------------------------------|-----------------|---------------------------|--------------|----------|---|---|-----------------|------|------------|-----------------|-----|-------------|--------------|----|----|--|
|                                 |                 | 0                         | 1            | 2        | 3 | 4 | 5               | 6    | 7          | 8               | 9   | 10          | 11           | 12 | 13 |  |
| Active (activate)<br>Open a row | L               | L                         | L            | Row R0–3 |   |   | Bank            |      | Bank group |                 |     | Chip CID0–2 |              |    |    |  |
|                                 | H               | Row R4–16                 |              |          |   |   |                 |      |            |                 |     |             | R17/<br>CID3 |    |    |  |
| Unassigned, reserved            | L               | L                         | H            | V        |   |   |                 |      |            |                 |     |             |              |    |    |  |
|                                 | H               | V                         |              |          |   |   |                 |      |            |                 |     |             |              |    |    |  |
| Unassigned, reserved            | L               | H                         | L            | L        | L | V |                 |      |            |                 |     |             |              |    |    |  |
|                                 | H               | V                         |              |          |   |   |                 |      |            |                 |     |             |              |    |    |  |
| Write pattern                   | L               | H                         | L            | L        | H | L | H               | Bank | Bank group |                 |     | Chip CID0–2 |              |    |    |  |
|                                 | H               | V                         | Column C3–10 |          |   |   |                 |      | V          | $\overline{AP}$ | H   | V           | CID3         |    |    |  |
| Unassigned, reserved            | L               | H                         | L            | L        | H | H | V               |      |            |                 |     |             |              |    |    |  |
|                                 | H               | V                         |              |          |   |   |                 |      |            |                 |     |             |              |    |    |  |
| Mode register write             | L               | H                         | L            | H        | L | L | Address MRA0–7  |      |            |                 |     | V           |              |    |    |  |
|                                 | H               | Data MRD0–7               |              |          |   |   |                 |      | V          | CW              | V   |             |              |    |    |  |
| Mode register read              | L               | H                         | L            | H        | L | H | Address MRA0–7  |      |            |                 |     | V           |              |    |    |  |
|                                 | H               | V                         |              |          |   |   |                 |      |            | CW              |     | V           |              |    |    |  |
| Write                           | L               | H                         | L            | H        | H | L | $\overline{BL}$ | Bank | Bank group |                 |     | Chip CID0–2 |              |    |    |  |
|                                 | H               | V                         | Column C3–10 |          |   |   |                 |      | V          | $\overline{AP}$ | WRP | V           | CID3         |    |    |  |
| Read                            | L               | H                         | L            | H        | H | H | $\overline{BL}$ | Bank | Bank group |                 |     | Chip CID0–2 |              |    |    |  |
|                                 | H               | V                         | Column C3–10 |          |   |   |                 |      | V          | $\overline{AP}$ | V   |             | CID3         |    |    |  |
| Vref CA                         | L               | H                         | H            | L        | L | L | Data            |      |            |                 |     |             |              |    | V  |  |

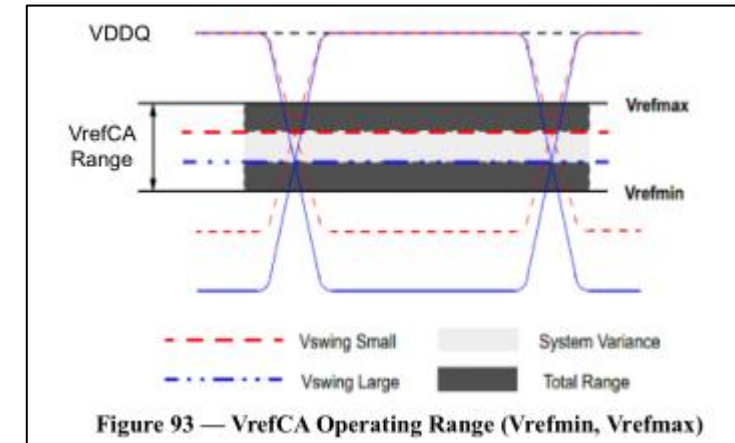
Data table from Wikipedia:  
[https://en.wikipedia.org/wiki/DDR5\\_SDRAM](https://en.wikipedia.org/wiki/DDR5_SDRAM)

# CA Bus Internal VREF

- DDR4 used external VREFCA
- DDR5 has internally generated VREFCA
  - Identical to VREFDQ
- Internally adjustable via register commands
  - From 45% to 97.5% of VDDQ
  - 0.5% increments
- Reduced PCB design requirements
  - No VTT/VREF regulator needed

Table 1 — DDR4 SDRAM X4/8 Ballout using MO-207

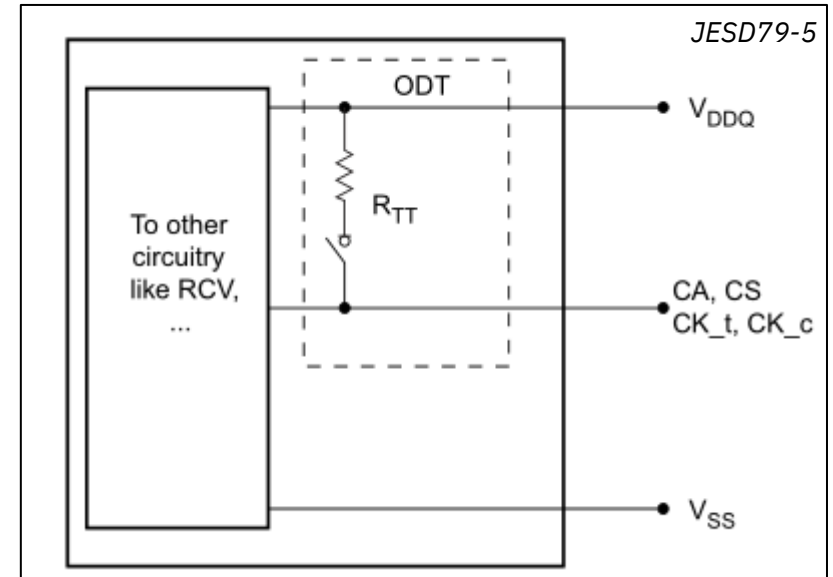
|   | 1       | 2                                      | 3                        | 4 | 5 | 6 | 7                                                         | 8                                         | 9                        |   |
|---|---------|----------------------------------------|--------------------------|---|---|---|-----------------------------------------------------------|-------------------------------------------|--------------------------|---|
| A | VDD     | VSSQ                                   | TDQS_c <sup>3</sup>      |   |   |   | DM_n, DBI_n<br>TDQS_t <sup>2</sup> ,<br>(NC) <sup>1</sup> | VSSQ                                      | VSS                      | A |
| B | VPP     | VDDQ                                   | DQS_c                    |   |   |   | DQ1                                                       | VDDQ                                      | ZQ                       | B |
| C | VDDQ    | DQ0                                    | DQS_t                    |   |   |   | VDD                                                       | VSS                                       | VDDQ                     | C |
| D | VSSQ    | DQ4<br>(NC) <sup>1</sup>               | DQ2                      |   |   |   | DQ3                                                       | DQ5<br>(NC) <sup>1</sup>                  | VSSQ                     | D |
| E | VSS     | VDDQ                                   | DQ6<br>(NC) <sup>1</sup> |   |   |   | DQ7<br>(NC) <sup>1</sup>                                  | VDDQ                                      | VSS                      | E |
| F | VDD     | (C2) <sup>5</sup><br>ODT1 <sup>6</sup> | ODT                      |   |   |   | CK_t                                                      | CK_c                                      | VDD                      | F |
| G | VSS     | (C0) <sup>5</sup><br>CKE1 <sup>6</sup> | CKE                      |   |   |   | CS_n                                                      | (C1) <sup>5</sup><br>(CS1_n) <sup>6</sup> | TEN<br>(NC) <sup>7</sup> | G |
| H | VDD     | WE_n<br>A14                            | ACT_n                    |   |   |   | CAS_n<br>A15                                              | RAS_n<br>A16                              | VSS                      | H |
| J | VREFCA  | BG0                                    | A10<br>AP                |   |   |   | A12<br>BC_n                                               | BG1                                       | VDD                      | J |
| K | VSS     | BA0                                    | A4                       |   |   |   | A3                                                        | BA1                                       | VSS                      | K |
| L | RESET_n | A6                                     | A0                       |   |   |   | A1                                                        | A5                                        | ALERT_n                  | L |
| M | VDD     | A8                                     | A2                       |   |   |   | A9                                                        | A7                                        | VPP                      | M |
| N | VSS     | A11                                    | PAR                      |   |   |   | A17<br>(NC) <sup>4</sup>                                  | A13                                       | VDD                      | N |



JESD79-5

# On-Die Termination Changes

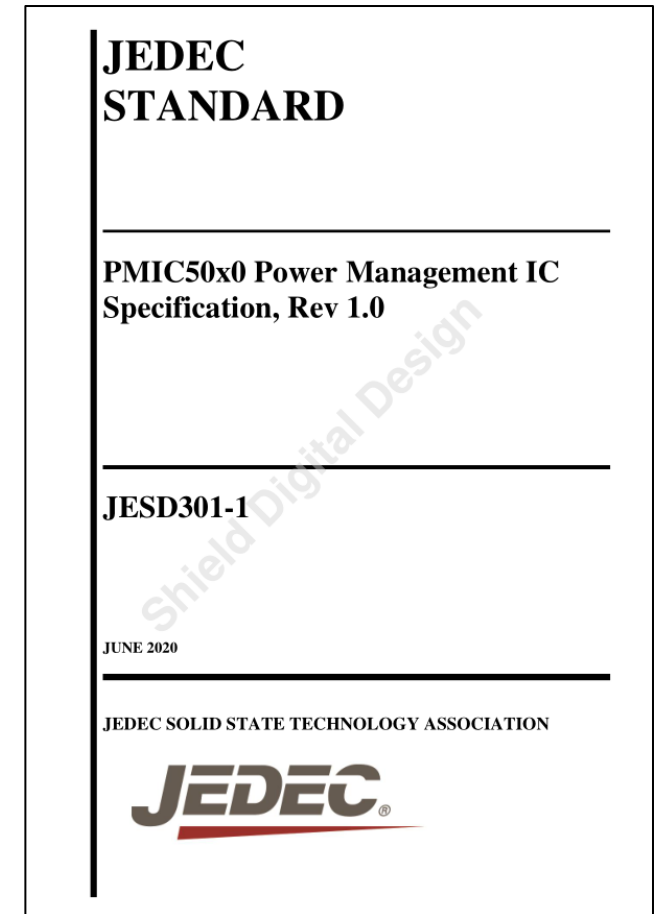
- All ODT is now fully command/register based
  - No dedicated ODT pin
- ODT is now available for CA bus
  - DQ/DQS still use direct routing from controller to DRAM
  - CA bus uses fly-by routing
- Pin CA\_ODT allows different settings for end-of-group
  - Could use  $40\Omega$  for end device but ODT disabled for other devices



## 2. New Features in DDR5

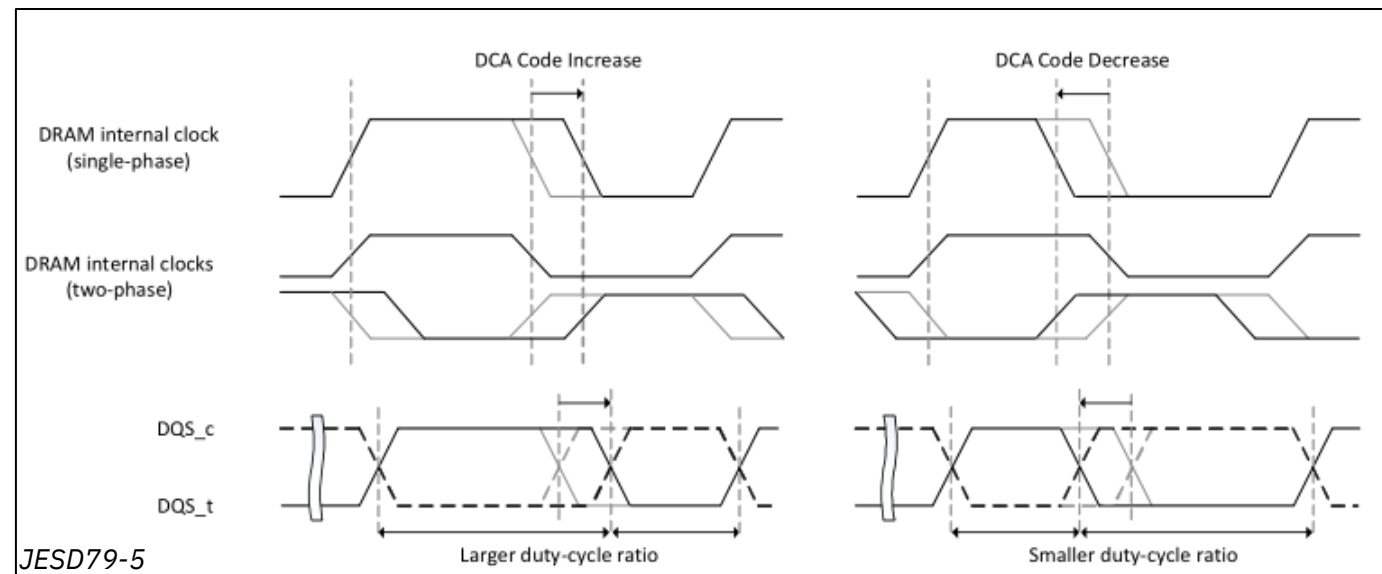
# VRM on DIMM

- Excellent for **power integrity**
  - DC (IR drop) and AC (PDN impedance)
- **Four SMPS, three LDO**
  - 15W of power
  - 4.25 – 15V input for SMPS, 3 – 3.6V input for LDO
  - OV protection for SMPS input
  - OV/UV/OC protection for SMPS output
  - ADC for voltage, current, temp measurement
- **Two management** interfaces
  - I2C or “I3C Basic”
    - I3C is a MIPI modification of I2C (open-drain or push-pull, faster data rates)
    - Same pins for both



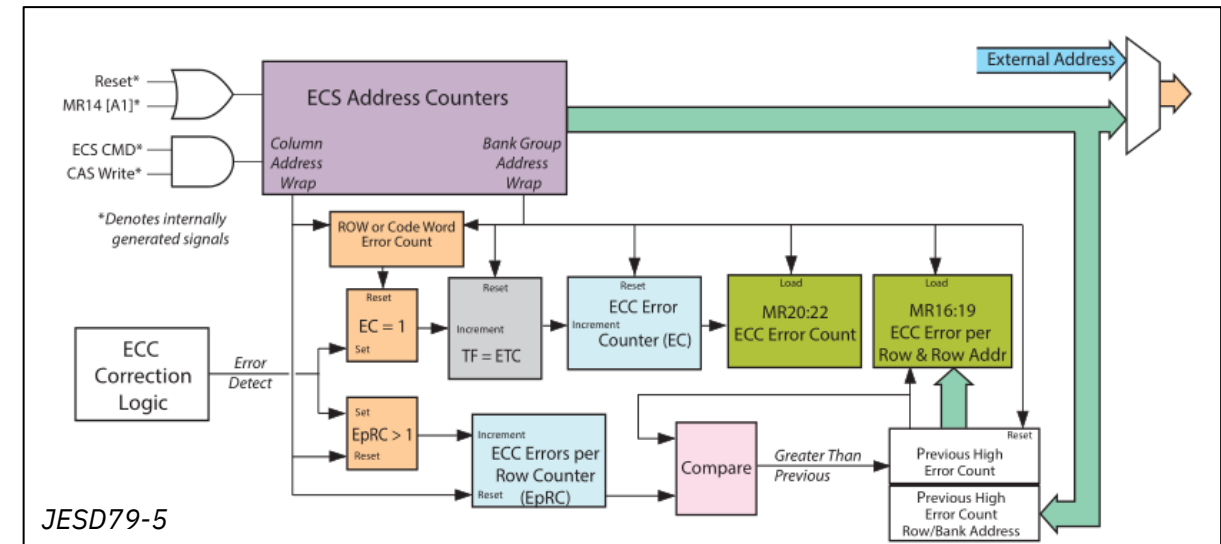
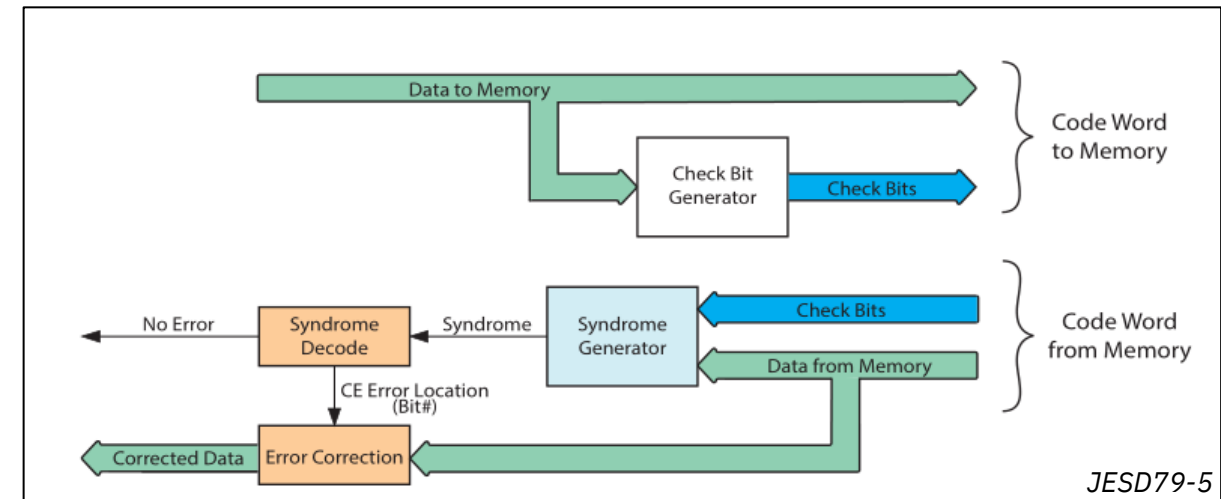
# Duty Cycle Adjuster

- Memory controller can **adjust** DRAM internal DQS/DQ duty cycle
  - Compensate for duty cycle error of all DQS/DQ
- Register-adjustable in **8-steps each**, positive and negative
- 28 to 56 ps adjustment **range**
- **Global or per-pin** adjustment



# On-Die ECC

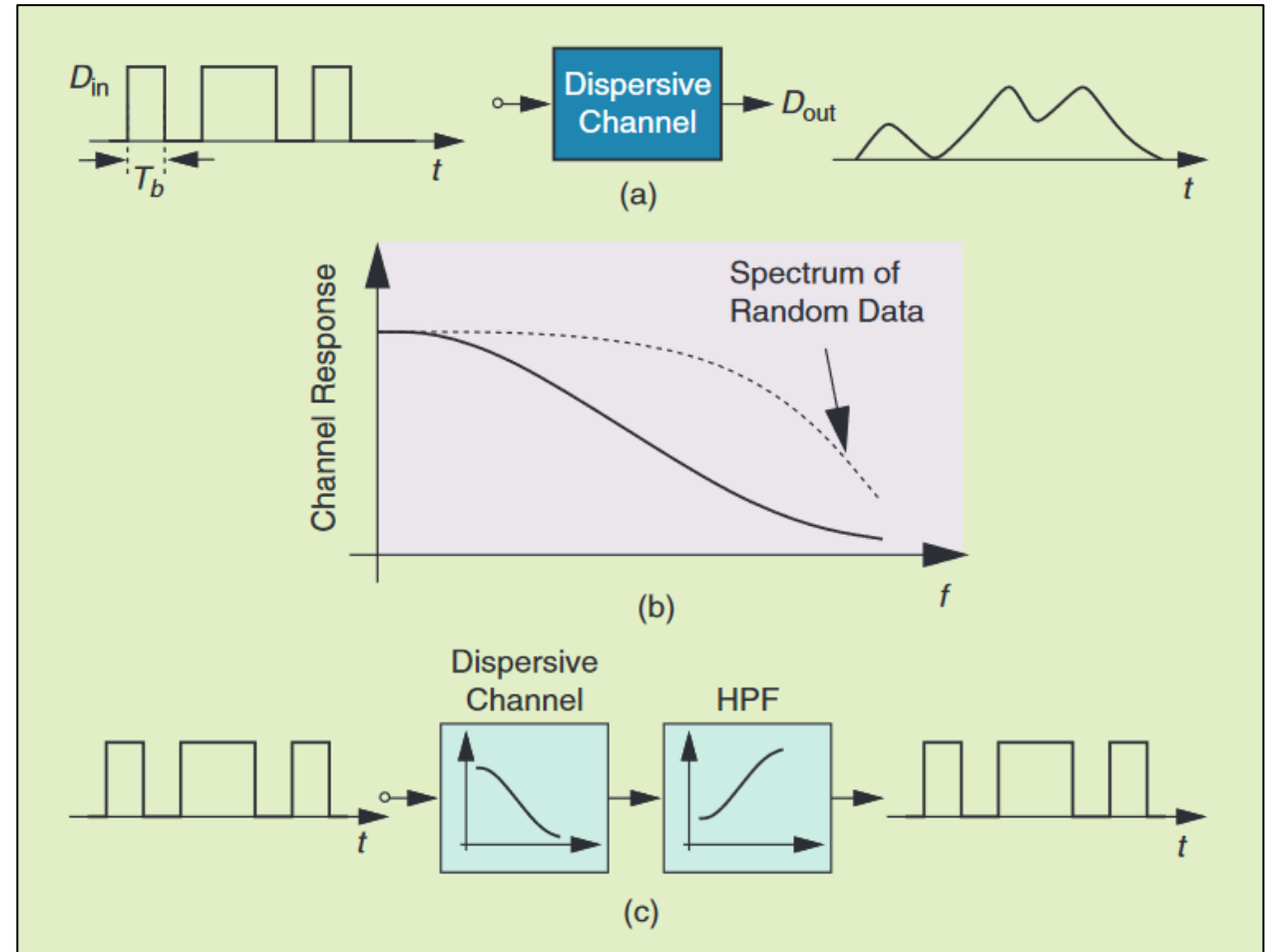
- Single Error **Correction**
- 128 data bits, 8 ECC check bits
- DRAM **dynamically corrects** single-bit errors
- Error **counting**/tracking



ECC Transparency and Error Scrub Block Diagram

# Decision Feedback Equalization

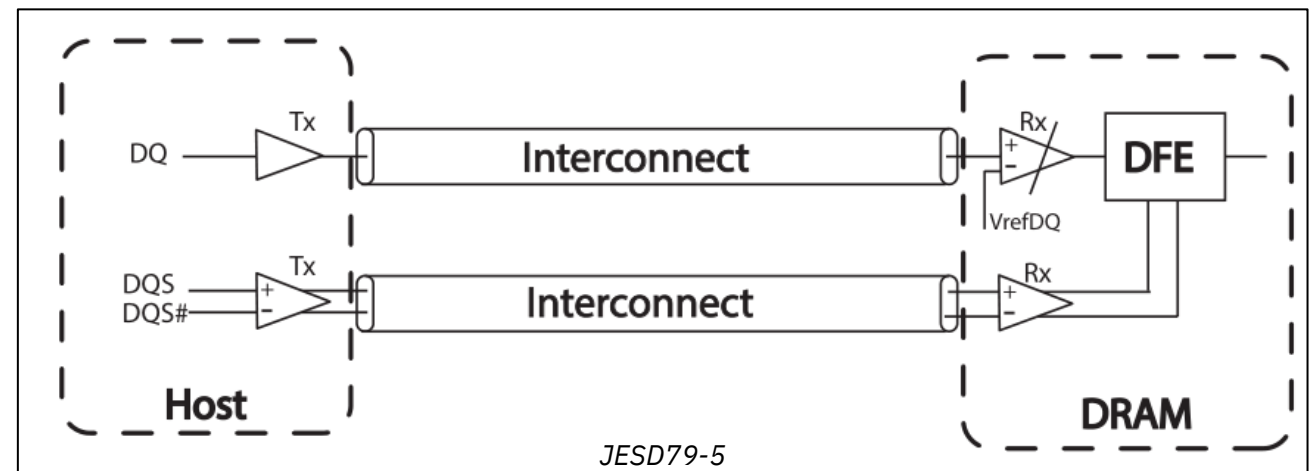
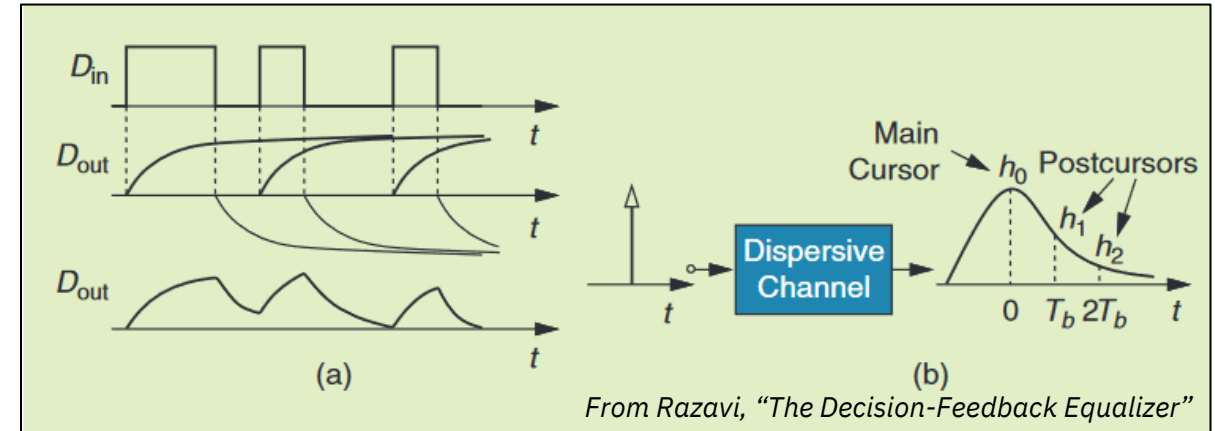
- **ISI** increases and eye closes
  - Due to higher data rates (at or above 3200 MT/s)
- Memory channels are **reflective**
  - Due to many impedance discontinuities
- Provides a high frequency **boost**
  - Compensates for channel loss



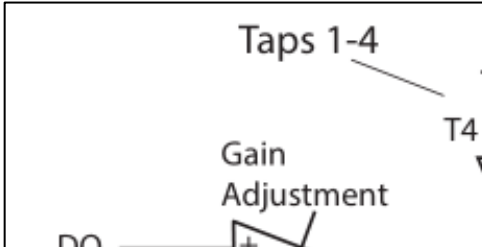
From Razavi, "The Decision-Feedback Equalizer"

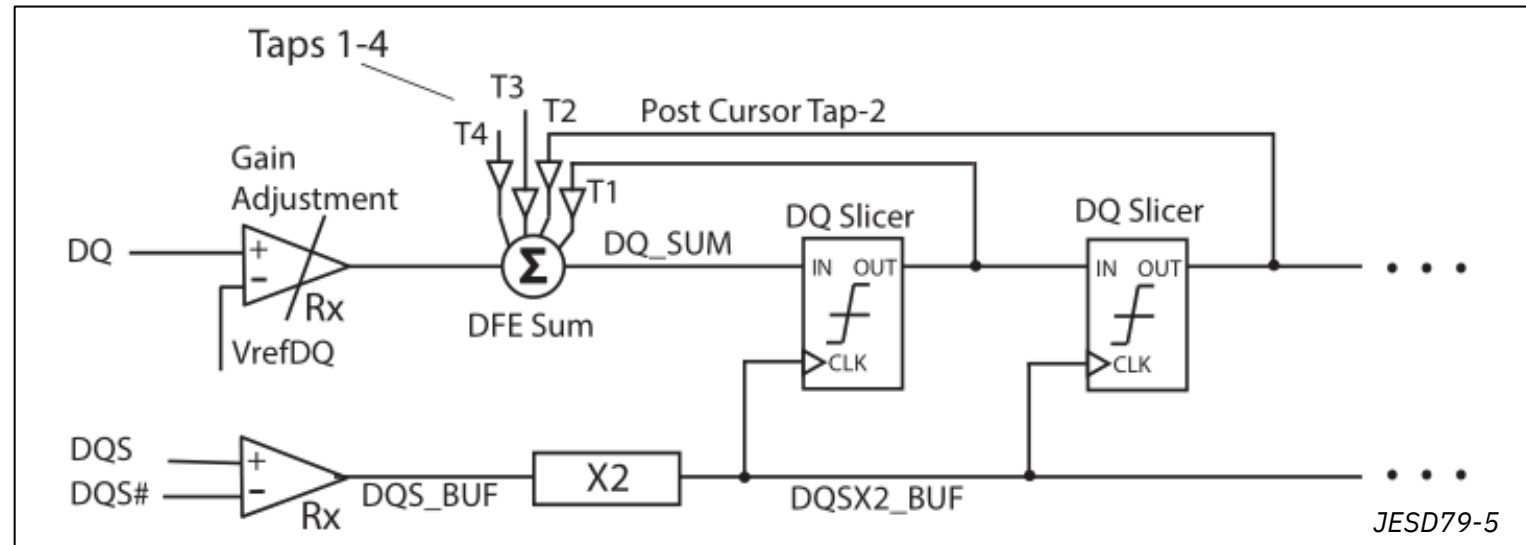
# Decision Feedback Equalization

- Approximate **linear equalization** with FIR filter
  - Only **linear** stages (delay and scaling)
- Typical pros/cons of DFE
  - No amplification of **noise** (such as CTLE)
  - **No pre-cursor** equalization
- Remove **scaled interference** from four previous bits
  - Based on the channel impulse response
  - Four bits in a 4-tap DFE



# Decision Feedback Equalization

- **4-Tap DFE** equalizes DQ signals
  - May implement 1-way, 2-way or 4-way **interleaved** DFE
    - 1-way shown below
    - Interleaving reduces slicer clock rate (and power)
  - Gain Bias:
    - **-6 dB to +6 dB**
- 
- The diagram shows a block labeled 'DQ' at the bottom left. A line from this block goes into a trapezoidal block labeled 'Gain Adjustment'. Above the 'Gain Adjustment' block, there are two labels: 'Taps 1-4' and 'T4', with lines pointing to the top of the block. The output of the 'Gain Adjustment' block goes into a block labeled 'Slicer'. The output of the 'Slicer' block is labeled 'DQ' at the bottom right.



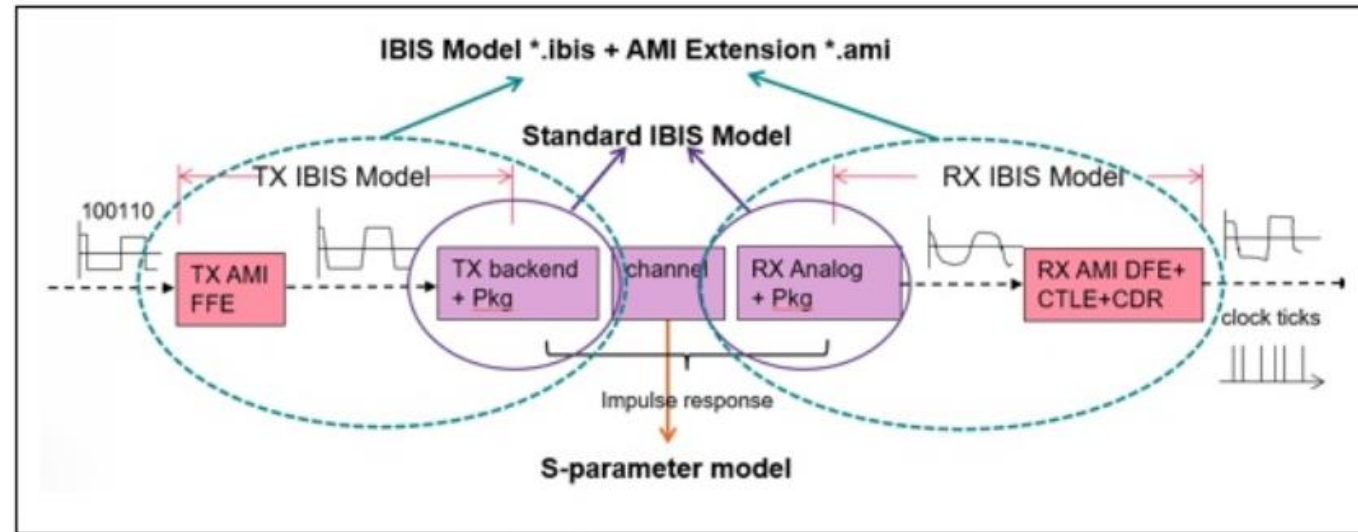
# Bringing It All Together: Simulation

- DDR5 made things better with DFE
  - But with increased data rate, **simulation is a must** to ensure successful system design
- Questions arise:
  - Advanced processing is on-chip with DFE, how do we simulate this?
  - Is a vendor AMI model required?
  - Will my power distribution network design withstand the faster switching speed?
- In order to answer the question: “**How do you know it will work?**”
  - We will incorporate both IBIS-AMI and power-aware simulation

# 3. DDR5 Simulation

# What is IBIS-AMI?

- Algorithmic Modeling Interface
- Extends traditional IBIS (behavioral models)
  - Equalization and other non-LTI effects
  - Ultra-low BER (statistical simulation)
  - Package sNp models
- Statistical simulation based on channel pulse response, or
- Time-domain simulation



<https://www.signalintegrityjournal.com/articles/2020-back-to-basics-ibisibis-ami-and-the-path-to-lpddr5>

# Demo: Plan

- Simulate one byte lane (DQ + DQS)
  - 6400 MT/s
  - 10" PCB trace model
  - DFE on the receiver
  - Verify minimum eye height/width meets spec (70mV/0.25UI at BER > 1E-16)

# Demo: Setup and PCB

- Prep: Have IBIS model in a clean directory ready to use
- Open TopXplorer, start a new topology
  - Menu: Topology->New
- Create a SystemSI/Parallel Bus Analysis/Blank
- Add CTRL IBIS, MEM IBIS
- Add Trace Model
  - Edit Trace, Microstrip:
    - 10x, 4mil width, 4mil space, 0.7mil thickness, 4mil substrate
      - System has extremely tight density requirements and cannot use up more area
    - Length: 10000mil
    - Include Power
  - Edit VRM: Voltage range: 1.1 V Typical; 1.067 to 1.166 V

# Demo: Controller IBIS

- Load IBIS
  - Select component “DDR5\_Controller”
  - Check “Explicit IO Power/Ground Terminals”
  - Verify the Bus Definitions
- Select CTRL IBIS
  - Uncheck “Ideal Power”
  - Check “Package Parasitics”
- Redo same steps for receiver (Don’t forget to select receiver component)

# Demo: Connections

- Controller side:
  - Verify power and grounds connected
  - Select DQ0 to DQ7 and A-H, right click and “Connect by pin pair”
  - Same with DQS0 P/N and I/J
- Follow same steps between trace and receiver

# Demo: Simulation Setup

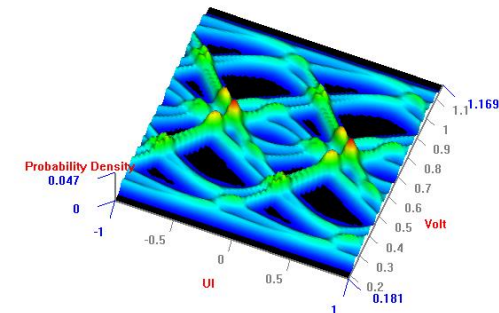
- Check “Use Channel Simulator”
- Check Connectivity, verify magnitude is ~0.8
- Set timing budget
  - Set data rate to 6.4 Gbps
- Set Analysis Options
  - Corner – Keep only Typ selected
  - Bus Simulation – select write only
  - Channel Sim – No change
  - IO Models:
    - Change stimulus to PRBS32
    - Verify DQS still 01/10
  - Change bits to display: Last 1000
  - Change jitter to include both RJ and DJ
  - Check only DQ0 and DQS P/N (Too long to run simulate all)

# Demo: Simulate

- Save Topology
- Enable Distributed and Set up Resources (Default)
- Run (Note: takes 2.5min on my machine)
- **Poll Question**

# Demo: Simulate

- Save Topology
- Enable Distributed and Set up Resources (Default)
- Run (Note: takes 2.5min on my machine)
- Eye is closed
- NOTE: Mention that this configuration wouldn't have worked with DDR4, but with the new addition of DFE to DDR5 we can make it work at even higher data rates

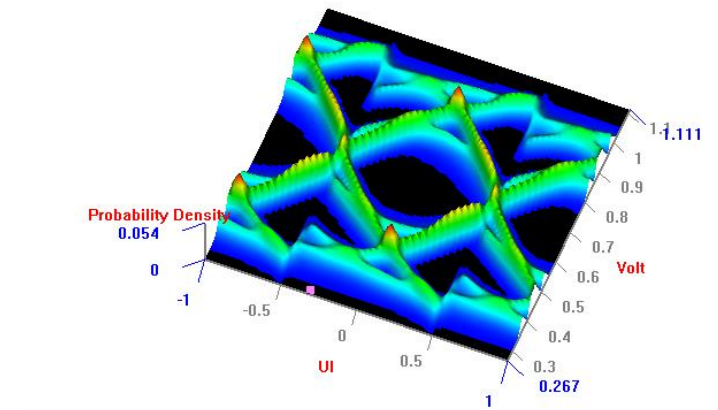


# Demo: Add DFE

- Right-click the MEM IBIS and choose “Add AMI”
  - Select the included “Memory Licensed” AMI/DLL files
  - Review the DFE tap settings (4-tap for DDR5)
  - Re-run
- 
- **Poll Question:**

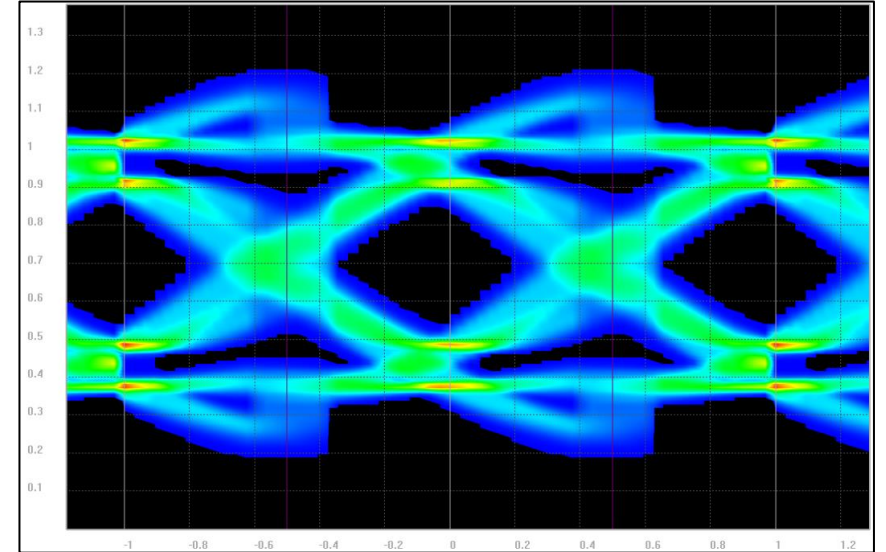
# Demo: Add DFE

- Right-click the MEM IBIS and choose “Add AMI”
- Select the included “Memory Licensed” AMI/DLL files
- Review the DFE tap settings (4-tap for DDR5)
- Re-run
- Results in a wide-open eye (0.54UI/140mV at BER of 1E-16)
  - Meets our requirements



# Conclusion

- DDR5 is a **major** performance improvement
  - Speeds from 3200 MT/s up to 8400 MT/s
- More **complex** simulation than DDR4
- Tools exist which can ensure **successful** design of your DDR5 system



# References

## 1. JEDEC JESD 79-5, DDR5 Specification

1. <https://www.jedec.org/system/files/docs/JESD79-5.pdf>

## 2. JEDEC JESD 79-4, DDR4 Specification

1. <https://www.jedec.org/system/files/docs/JESD79-4.pdf>

## 3. JEDEC JESD 301-1 PMIC Specification

1. <https://www.jedec.org/system/files/docs/JESD301-1.pdf>

## 4. Micron DDR5: Key Module Features

1. [https://media-www.micron.com/-/media/client/global/documents/products/technical-marketing-brief/ddr5\\_key\\_module\\_features\\_tech\\_brief.pdf](https://media-www.micron.com/-/media/client/global/documents/products/technical-marketing-brief/ddr5_key_module_features_tech_brief.pdf)

## 5. Razavi, *The Decision-Feedback Equalizer*

1. <https://www.seas.ucla.edu/brweb/papers/Journals/BRFall17DFE.pdf>

# Thank you!